

How Cities Adopt Networked Infrastructure

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Summary

Gas, electricity and internet each spread through cities in the same order: a public or institutional anchor first, then commerce, then affluent households, then everyone else once pricing, regulation or subsidy changed the economics.

Each network needed three things before it could serve a home: a central source, a legal right to run lines through public streets, and equipment inside the building. Each also leaned on what came before. Gas companies won street access by lighting public lamps, electric utilities grew on transit load, and broadband ran over telephone and cable television wires.

Who came first was decided by revenue per length of line. Dense, wealthy and commercial districts were served early. Low-income blocks and rural areas were reached by a separate device in each case: the prepayment meter for gas, federal loans for rural electricity, and franchise enforcement and federal grants for broadband. That last step is the least dependable. It arrived under regulated monopoly between about 1850 and 1960, and it has not arrived in the same way for broadband.

The sections below give the evidence for each network, the shared mechanisms, and how networks expand once built. The last two sections list what is still unsourced and which works would firm it up.

Gas lighting (1810s onward)

Gas reached streets and shops first and ordinary homes last. Street-lighting contracts were the price of permission to dig, and private supply went only to blocks that could pay.

What supply required

- **A central works and a pipe network.** Boulton and Watt developed gas plants from 1802 and sold them mostly to individual mills. London's Gas Light and Coke Company first planned many small works, each feeding 100 to 150 nearby lamps, then moved to a few large stations as customers grew ([Tomory 2011, pp. 77, 79](#)).

- **A charter and the right to open the streets.** Parliament's 1810 act gave the company street-opening rights, and its charter followed in 1812. In return it had to light streets for less than oil lamps cost. It made no profit on those contracts and pursued them as the easiest route to permission to lay pipe ([Tomory 2011, pp. 81-82](#)). Chicago's 1850 franchise paired a ten-year monopoly on pipes with a city contract for street lamps (Platt 1991, p. 12).
- **Control over fittings and use.** The London company was barred from installing pipes inside buildings. It certified outside fitters, standardized parts, and sent inspectors to enforce burning hours on flat-rate customers ([Tomory 2011, pp. 78, 97-100](#)). The rotary meter dates from 1817, and domestic meters became common only after 1850 ([National Gas Museum](#)).
- **Pressure management as the network spread.** Gas rises, so higher districts drew more than their share. The company isolated them with regulators, and linked the mains of its three works so supply could shift between them ([Tomory 2011, pp. 85, 96](#)).

Who was served first

- **Public buildings and street authorities, then shops and homes.** In mid-1814 the London company had four customers: two street-lighting authorities, Westminster Hall and a church. By 1816 it had about 2,400 customers and 8,600 lamps. By 1820 it supplied about 30,000 lamps over 122 miles of mains ([Tomory 2011, pp. 91, 101](#)).
- **Commerce along the first mains.** Chicago's company signed 125 shopkeepers on and near Lake Street before it made any gas, and its mains grew from 5 to 53 miles by 1860 (Platt 1991, pp. 12-13). Baltimore chartered the first US gas company in 1816 ([WTOP 2016](#)). New Haven lit street lamps in 1848 and served 1,252 homes and businesses by the early 1850s ([Howe](#)).
- **Only the well-to-do at home.** Chicago's companies generally did not extend into working-class neighborhoods and had no legal duty to. They held the price at \$3 per thousand cubic feet for over 25 years, and would not lay mains without near-certainty that every building on a block would connect (Platt 1991, p. 14).
- **Poorer households decades later.** The prepayment or slot meter, introduced after 1870, put gas within reach of poorer people ([National Gas Museum](#)).
- **Territory was carved up, not contested.** Chicago's two gas companies secretly split the city along the river (Platt 1991, pp. 13-14).

Electricity (1880s onward)

Electricity began in financial districts and on transit lines, reached about seven in ten US homes by 1930, and reached farms only after federal lending began in 1935.

What supply required

- **A station inside the district it served.** Edison's Pearl Street station opened on 4 September 1882 and supplied a quarter of a square mile of lower Manhattan ([ETHW, Pearl Street Station](#)). In Chicago the price of copper put a practical limit of about a half-mile radius on a direct-current station. The first underground network there cost \$197,100 for 19 miles, against \$113,900 for the station itself (Platt 1991, pp. 33, 71).
- **A municipal franchise.** Chicago required underground wiring in the centre, so rivals sought franchises that allowed cheaper overhead wires on the fringe. In 1897–98 Insull acquired a fifty-year franchise that allowed overhead wires everywhere outside the centre (Platt 1991, pp. 70, 81–82). A state utility commission followed in 1913 ([Encyclopedia of Chicago](#)).
- **Large customers to carry the fixed cost.** In 1902 Chicago's street and elevated railways used over three times the energy of all Edison's light and power customers. Insull signed them at very low rates, and the utility's load factor rose from 30 to 40 percent between 1898 and 1908 (Platt 1991, pp. 119, 122). Their guaranteed contracts let him borrow to build (Platt 1991, p. 127).
- **Rates that made small customers worth serving.** Two-part rates introduced in 1898 cut the average residential price from 19.5 to 13.33 cents per kilowatt-hour, and the company wired six outlets free (Platt 1991, p. 89).

Who was served first

- **Luxury hotels, stores and offices downtown.** Chicago's first arc lamps went into the Grand Pacific Hotel and the Palmer House in 1880. By 1885 the city had over a thousand arc lamps, almost all in the central business district (Platt 1991, pp. 22, 29). Pearl Street's early customers included the New York Times and Drexel, Morgan and Company ([ETHW](#)). Early utilities put their generating plants near the elite (Nye 1990, p. 242).
- **The rich, by private plant.** From 1881 to 1887 the Edison system in Chicago was sold only as self-contained plants, about 30,000 lamps in all, to theaters, banks, department stores, factories and a few mansions. In 1892 the city had 18 central stations and 498 self-contained plants (Platt 1991, pp. 33–37, 68). Nationally, towns built or franchised their own street lighting, and many streetcar lines and factories ran their own plants (Nye 1990, p. 139).

- **Homes came late because of the load curve.** Street and commercial lighting peaked in the evening, so adding homes meant adding generating plant. Once street railways and factories moved the peak to daytime, homes could be served from plant already built. By 1910 the lines serving commerce and industry already ran into most sections of American cities. Until then the home market was about 5 percent of households (Nye 1990, pp. 259–260).
- **Then homes filled in fast, in cities.** One US home in seven had electricity in 1910 and seven in ten in 1930. Of the newly wired homes, 85 percent were urban or suburban (Nye 1990, p. 261). Chicago's residential Edison customers rose from fewer than 5,000 in 1902 to almost 80,000 in 1912, about 16 percent of families, and to over a third of households by the eve of the First World War (Platt 1991, pp. 147, 160).
- **Working-class districts met electricity in public first.** Transit lines, lit shopping strips and five-cent theaters arrived while the homes around them stayed on gas. By 1912 Insull offered builders of workers' housing a \$12 wiring package, against \$100 to \$300 for an affluent home (Platt 1991, pp. 144, 147, 150–151).
- **Homes paid the highest rates but were small accounts.** General Electric reported that residential customers used about 30 percent of the energy in 1922 and paid about half of utility income (Nye 1990, p. 261). Tobey gives domestic customers as 82.7 percent of users and 29.4 percent of revenue in 1929 ([Tobey 1996](#)). The two revenue shares do not agree.
- **Farms waited for the state.** Two percent of US farms had electricity in 1910 and one in nine in 1935. Utilities judged that little more than 10 percent of farms could be served at a profit, and those at higher rates and lower returns than dense suburbs (Nye 1990, pp. 288, 299–300). The Rural Electrification Administration was created on 11 May 1935, and the figure was almost 97 percent by 1960 ([Living New Deal](#)).

Internet and broadband (1980s onward)

The internet started in research institutions and reached homes over wires built for telephone and television, so upgrades went first to the neighborhoods where carriers expected the best return.

What supply required

- **A publicly funded backbone.** NSFNET began operating in 1986, linking five supercomputing centers under a policy that limited it to research and education ([Wikipedia, NSFNET](#)). On 30 April 1995 the government shut the backbone and handed the infrastructure to private companies (Mack 2020, p. 63).

- **Existing last-mile wires.** Broadband rollout is local and depends on each provider's use of existing infrastructure. DSL, offered from 1995, runs over telephone lines, and cable broadband runs over cable television plant and reaches farther from its hub (Mack 2020, pp. 63–64).
- **Existing long-distance corridors.** A map of US long-haul fibre found 542 shared conduits. A significant share run alongside roads, fewer along railways, and some along pipelines (Durairajan 2020, pp. 52–54).
- **Proximity to the exchange.** A DSL user has to be within a few miles of a telephone switching office, and the signal weakens even inside that range ([Britannica, DSL](#)).
- **A franchise, with build-out terms.** New York City's cable franchise with Verizon carried build-out obligations. The city's 2020 settlement required service to 500,000 more households, with priority for public housing and 11 low-income community districts ([Smith 2020, CityLand](#)).

Who was served first

- **Universities and labs, then business districts.** Around 2000 the City of London had at least six separate fibre-optic grids beneath its financial district. In mid-1999 about 86 percent of US internet delivery capacity sat in the prosperous suburbs and business areas of the twenty largest cities (Graham and Marvin 2001, pp. 318, 375).
- **Affluent neighborhoods next, by design.** Fibre access networks then cost \$6,000 to \$10,000 per home passed, so providers deployed them first only in more affluent areas. AT&T reorganized its marketing after finding that 80 percent of its profit came from 20 percent of its customers (Graham and Marvin 2001, pp. 236, 251).
- **Households that could pay.** Home broadband rose from 1 percent of US adults in 2000 to a majority by 2007 and 80 percent in 2023 ([Pew Research Center, fact sheet](#)). In early 2021, 43 percent of adults with household incomes under \$30,000 still had no home broadband ([Vogels 2021, Pew](#)).
- **Cities before the countryside, and not only by income.** A review of the research finds an urban bias in broadband provision and a large literature on urban and rural gaps. One study it cites found racial differences in DSL adoption even after controlling for income and education (Mack 2020, pp. 68, 70).
- **Suburbs before poor city blocks.** AT&T's upgraded VDSL or fiber service reached 61 percent of suburban Cuyahoga County census blocks but 34 percent of Cleveland blocks, according to 2016 FCC filings ([Callahan 2017, NDIA](#)).

- **Demand-led rollout followed old lines.** Google Fiber built first in Kansas City neighborhoods that pre-registered enough households at \$10 each. Wealthier, mostly white areas west of Troost Avenue qualified, and lower-income, mostly Black areas to the east did not ([Halegoua 2014](#)).
- **Public money for the remainder.** The \$42.45 billion Broadband Equity, Access and Deployment program funds infrastructure in unserved and underserved areas ([NTIA](#)).

The shared pattern

Six mechanisms recur across the three networks, and each is documented in the sources cited above.

Mechanism	Gas	Electricity	Internet
Rides an existing right-of-way	Mains under public streets, modelled on water companies	Conductors under the same streets, laid out on the gasworks model	Telephone and cable lines to the home; roads, rail and pipelines for long-haul fibre
An anchor customer comes before homes	Street-lighting contracts and public buildings	Hotels and offices, then streetcars and elevated railways	Research and education institutions
Revenue per customer sets the order	No mains unless a whole block would connect	Homes were 82.7% of customers and 29.4% of revenue in 1929	Upgrades reached 61% of suburban blocks and 34% of city blocks in Cuyahoga County
A distance limit draws the first map	Each early works was meant to feed 100 to 150 nearby lamps	About a half-mile radius per direct-current station	DSL only within a few miles of an exchange
Franchise leads to monopoly, then regulation	Exclusive franchises; a secret territorial split in Chicago	Rival franchises bought out by 1898; state commission in 1913	Cable franchise with build-out terms, enforced by lawsuit in New York

Mechanism	Gas	Electricity	Internet
A separate device reaches the last users	Prepayment slot meter, after 1870	Two-part rates in the city; federal loans for farms, 1935	Settlement priority districts; federal grants for unserved areas

Platt states the spatial rule directly for Chicago: gas, water and streetcars were all planted first at the city centre, where density and wealth were greatest, and grew outward from there (Platt 1991, p. 25). Edison copied the gasworks model on purpose, down to calling his conductors mains (Platt 1991, p. 32; [Tomory 2011, p. 102](#)).

The order of service in every case was:

1. Public or institutional anchor (street lamps, transit, universities).
2. Commercial premises along the first lines.
3. Affluent households near those lines.
4. Ordinary urban households, once pricing or metering changed.
5. Low-density and low-income areas, by mandate or subsidy.

Graham and Marvin place this sequence in a longer arc. From about 1850 to 1960, scattered islands of infrastructure were joined into standardized, regulated monopolies, and cross-subsidies and universal service obligations carried them into homes and rural areas (Graham and Marvin 2001, pp. 40–41, 81). They argue that liberalization since then has been undoing step 5: new networks are built to connect valued users and places and to bypass the rest (pp. 167–168). On that reading, universal service belongs to one historical period and does not follow automatically from the earlier steps.

Expanding the network: stations and substations

New generating stations add capacity, while substations and feeders extend reach, and after about 1900 the two were sited by different rules.

Where

- **Under direct current, a station per district.** Before 1897 Chicago Edison ran four generating stations: two large ones near the electrical centre of the system on the river, and two small ones in outlying residential districts beyond their economical

reach. Its longest feeders ran about 6,600 feet, and distributing over even a one-mile radius took an excessive investment in copper ([Smith 1903, p. 209](#)). Manhattan had seven such stations by 1901 ([Blalock 2013](#)).

- **A local station only where load density paid for it.** A direct-current plant and network was practicable only in thickly settled territory with enough load inside the radius. Everywhere else, lighting came by 1,000-volt alternating-current circuits with a transformer at each house ([Smith 1903, p. 210](#)).
- **Substations went first into existing stations.** The first transmission, in 1897, ran about three miles at 4,500 volts from the Harrison Street station to the old 27th Street station, which kept feeding its existing street network. The stated aim was to reach outlying territory without duplicating investment or writing off the existing system ([Smith 1903, pp. 210–211](#)). In 1899 the first true substation went onto the North Side to carry the evening peak (Platt 1991, p. 81).
- **Then they followed load, space and the need for backup.** By 1903 Harrison Street fed fifteen substations over lines one to five miles long, and nearly every one had a duplicate line by a different route. Downtown substations sat in basements, where head room capped converter size at 500 kilowatts ([Smith 1903, pp. 212, 215–216](#)). There were 22 distribution points by 1902 and 33 by 1908 (Platt 1991, pp. 114–115). The White City amusement park got a substation of its own (Platt 1991, p. 146).
- **The big plant moved out to the water.** The Fisk Street station, started on 2 October 1903, sat on the river's south branch about two miles beyond the congested centre (Platt 1991, p. 114). The Adams plant at Niagara Falls, running from 1895, put generation at the resource itself ([ETHW, Adams Plant](#)).
- **Suburbs were bought and linked, not built fresh.** Insull bought the small utilities from Evanston to Waukegan, ran a high-voltage line between them with a string of substations, and tied the result to the city system with a 20,000-volt line in 1907 (Platt 1991, p. 173).
- **Today, where the grid can accept it.** About 2,060 gigawatts of generation and storage were waiting for a grid connection at the end of 2025, with a median wait of over five years for projects built that year ([Lawrence Berkeley National Laboratory 2026](#)).

When

- **When existing plant is full.** In January 1901 Insull told his board that no new business could be accepted on Chicago's North Side, because generators and lines there were at capacity (Platt 1991, p. 109). By 1903 the Harrison Street station had reached its ultimate capacity and a 100,000-horsepower station was under construction ([Smith 1903, p. 212](#)).

- **When a large customer underwrites it.** Transit contracts with guaranteed minimum use let Insull borrow cheaply, so that commuters in effect financed the system's growth (Platt 1991, p. 127; [Platt interview, PBS](#)).
- **When bigger machines beat small stations on cost.** Fisk Street's first 5,000-kilowatt turbine was replaced within four years by 12,000-kilowatt units of the same size and cost, burning about a third less coal per unit (Platt 1991, p. 114).
- **When storage can no longer cover the peak.** Substation batteries were charged late at night and discharged at the afternoon peak. That saved generating plant only while the peak lasted under about two hours ([Smith 1903, p. 217](#)).
- **When expected peak demand outruns capacity.** Planners compare the reserve margin, capacity minus peak demand as a share of peak demand, with a regional target ([EIA 2012](#)).
- **When cost per mile is covered.** Private utilities quoted \$1,500 to \$2,000 per mile for rural lines and did not build them. Federally financed cooperatives were building for under \$825 per mile by 1939 ([Living New Deal](#)).
- **Gas followed the same logic earlier.** London's gas company sized new mains at four to eight times current load, and from 1815 mapped every main and valve to manage supply ([Tomory 2011, pp. 95–96](#)).

Smith's paper settles the siting question for 1903 practice. Four things decided where a substation went: enough load within feeder reach, an existing station or street network to reuse, two independent routes for the supply line, and physical room for the machines. He gives no numerical threshold. One discrepancy: Platt puts the first transmission at 2,300 volts and Smith, the company's own engineer, at 4,500.

Exceptions and limits

The pattern is a strong tendency in US and British cases. The gas and electricity evidence rests on three scholarly studies and one engineering paper from 1903. The internet evidence is thinner: one book from 2001, one research review from 2020, and otherwise survey data and secondary reports.

Where the pattern bends

- **The core is not always first.** In the Cleveland data, suburbs received the upgraded network ahead of the central city ([Callahan 2017](#)). Once a network exists, upgrades can run outside-in.

- **New building can lead old building.** By about 1890 builders on Chicago's affluent fringe had to wire new houses to compete. In 1910, 175 of 193 flats in one new North Side block had electric service, when even two years later only about one family in six did citywide. Working-class subdivisions were still sold without utility hookups (Platt 1991, pp. 70, 147, 150, 159).
- **Pricing can widen access without a mandate.** Insull concluded it was worth wiring residential districts at a short-term loss, because homes peaked two hours after shops and spread the load (Platt 1991, pp. 157–159). He also sent a truck with 10,000 electric irons through the neighborhoods as a free trial for new customers (Platt 1991, pp. 156–157).
- **How much the market delivered is disputed.** Tobey accepts that private companies had wired half of urban homes by the early 1920s. He argues that full electrical service for working-class households came from New Deal policy ([Tobey 1996](#)).
- **Universal service may be a phase, not an endpoint.** Graham and Marvin argue that the mid-century push to full coverage depended on monopoly and cross-subsidy, and that competitive networks now select their customers (2001, pp. 81, 167–168, 236).
- **A cheaper successor can reach the last users instead.** Warf reports more than ten times as many mobile users as landline users worldwide by about 2010. He concludes that mobiles delivered near-universal telephone service even while deregulation encouraged providers to pick their customers (2013, pp. 222, 227).
- **The device for the last users can also hold them at arm's length.** Over 3 million UK households had prepayment electricity meters by 1994. Graham and Marvin present these as a way for utilities to avoid dealing with poorer customers (2001, p. 298), where the 1870s slot meter is usually told as a story of widened access.

Limits of this draft

- **Geography.** Every case is from the United States or Britain, and the city-level electricity detail leans on Chicago.
- **Source quality.** London gas comes from Tomory's article. Chicago gas and electricity come from Platt's book and Smith's paper, and national electrification from Nye. The internet section has Graham and Marvin up to 2000 and a 2020 research review after that. It still has no study of rollout inside a city since 2001 apart from the Kansas City article.
- **Partial reading.** Tomory and Smith were read in full. In Platt I read chapters 2 and 6 in full and chapters 1, 4, 5 and 7 by targeted passage. In Nye I read about ten pages of chapters 6 and 7 plus keyword hits elsewhere. In Graham and Marvin I read about

fifteen pages located by keyword. In *Geographies of the Internet* I read chapter 5 and two pages of chapter 4.

- **Conflicting figures.** Pearl Street's first-day customers are about 85 in ETHW and 59 in the Census Bureau story. Chicago's first transmission voltage is 2,300 in Platt and 4,500 in Smith. Homes' share of utility revenue in the 1920s is about half in Nye and 29.4 percent in Tobey.
- **Two pages would not open.** Historic England's manufactured-gas history and the FCC's page on broadband connection types.

Claims from our earlier conversation, checked against the new sources

Earlier claim	Status	Evidence
Direct current reached about a mile	Supported, with a caveat	One mile was the working radius in Chicago, at excessive copper cost (Smith 1903, p. 209). Platt gives a half-mile as practical and a mile and a half as the economic maximum (1991, pp. 33, 72)
Tenants lagged because landlords controlled wiring	Not supported for Chicago	Flats were most new residential connections from 1909 to 1916 (Platt 1991, p. 160). Some landlords did ban early wiring as a hazard (p. 30)
New subdivisions get a mature technology before old cores	Partly supported	See "New building can lead old building" above
Gasworks sat on low ground near coal delivery	Partly supported	Gasholders were set low because gas rises (Tomory 2011, p. 96). A canal link to the coalfields made gas viable in Chicago (Platt 1991, p. 11)
Fisk Street undercut neighborhood stations	Supported	Platt 1991, pp. 112–115. Smith calls the small outlying stations inefficient (1903, p. 209)
Prepayment gas meters spread mainly in the 1890s	Unsourced	

Earlier claim	Status	Evidence
DSL reaches about three miles	Unsourced	Britannica says only "a few miles"
Niagara power reached Buffalo in 1896	Unsourced	
About 35 percent of US homes had electricity in 1920	Unsourced	

References

Opened for this draft

Source	Kind
Platt, H. L. (1991). <i>The Electric City: Energy and the Growth of the Chicago Area, 1880–1930</i> . University of Chicago Press. ACLS Humanities E-Book edition, 2016, open access.	Scholarly monograph, from company archives. Chapters 1, 2 and 4 to 7 used
Nye, D. E. (1990). <i>Electrifying America: Social Meanings of a New Technology, 1880–1940</i> . MIT Press.	Scholarly monograph. Chapters 6 and 7 used, plus single pages elsewhere
Tomory, L. (2011). Building the First Gas Network, 1812–1820. <i>Technology and Culture</i> 52(1): 75–102.	Peer-reviewed article, from company minute books. Read in full
Smith, E. F. (1903). The Development, Equipment, and Operation of the Sub-stations and Distributing Systems of the Chicago Edison Co. and Commonwealth Electric Co. <i>Journal of the Western Society of Engineers</i> 8: 209–222.	Primary source: engineering paper by a company engineer. Read in full, from scanned text
Graham, S. and Marvin, S. (2001). <i>Splintering Urbanism: Networked Infrastructures, Technological Mobilities and the Urban Condition</i> . Routledge.	Scholarly monograph. About fifteen pages used

Source	Kind
Mack, E. (2020). The history of broadband. In B. Warf, ed., <i>Geographies of the Internet</i> , pp. 63–76. Routledge.	Research review. Read in full
Durairajan, R. (2020). Robustness and the internet: a geographic fiber-optic infrastructure perspective. In B. Warf, ed., <i>Geographies of the Internet</i> , pp. 47–62. Routledge.	Book chapter reporting a 2015 mapping study. Two pages used
Warf, B. (2013). Geographies of global telephony in the age of the internet. <i>Geoforum</i> 45: 219–229.	Peer-reviewed article, global scale. Two points used
Blalock, T. J. (2013). The Rotary Era, Part 2: Early AC-to-DC Power Conversion, Continued. <i>IEEE Power and Energy Magazine</i>, Nov/Dec.	Engineering history
Britannica. DSL.	Reference
Callahan, B. (2017). AT&T's Digital Redlining of Cleveland. National Digital Inclusion Alliance.	Advocacy analysis of FCC Form 477 data
Encyclopedia of Chicago. Gas and Electricity.	Reference
Engineering and Technology History Wiki. Milestones: Pearl Street Station, 1882.	Reference
Engineering and Technology History Wiki. Milestones: Adams Hydroelectric Generating Plant, 1895.	Reference
Halegoua, G. R. (2014). Calling all "Fiberhoods": Google Fiber and the Politics of Visibility. <i>International Journal of Cultural Studies</i>. Author's manuscript.	Peer-reviewed scholarship
Howe, E. T. Early Connecticut Gas Light Companies. ConnecticutHistory.org.	Historian's essay
Lawrence Berkeley National Laboratory (2026). Queued Up: 2026 Edition.	Primary data

Source	Kind
Living New Deal. Rural Electrification Administration (1935).	Reference
National Gas Museum. Measuring gas.	Museum page
National Telecommunications and Information Administration. Broadband Equity, Access and Deployment Program.	Government program page
Pew Research Center (2025). Demographics of Internet and Home Broadband Usage in the United States.	Primary survey data
Platt, H. Interview for <i>Great Projects</i>. PBS.	Historian interview
Smith, L. (2020). City Reaches Settlement Agreement with Verizon FiOS Regarding Expansion. CityLand.	Legal news
Tobey, R. C. (1996). <i>Technology as Freedom: The New Deal and the Electrical Modernization of the American Home</i>. University of California Press.	Scholarly monograph
US Census Bureau (2026). History and the Census: Electricity.	Official statistics
US Energy Information Administration (2012). Reserve electric generating capacity helps keep the lights on.	Government explainer
Vogels, E. A. (2021). Digital divide persists even as Americans with lower incomes make gains in tech adoption. Pew Research Center.	Primary survey data
Wikipedia. National Science Foundation Network.	Reference
WTOP (2016). BGE turns 200.	Journalism

Standard scholarship, not opened

These are the works a reviewer would expect. Access to them would let me replace the weaker citations above.

- Grubestic, T. and Mack, E. (2015). *Broadband Telecommunications and Regional Development*. Routledge. Mack's chapter points to this for where and when broadband was deployed. It is the best lead for the internet section.
- Grubestic, T. and Murray, A. (2002). Constructing the divide: spatial disparities in broadband access. *Papers in Regional Science* 81(2): 197–221.
- Prieger, J. and Hu, W. M. (2008). The broadband digital divide and the nexus of race, competition, and quality. *Information Economics and Policy* 20(2): 150–167.
- Hughes, T. P. (1983). *Networks of Power: Electrification in Western Society, 1880–1930*. Johns Hopkins University Press. For cities other than Chicago.
- Rose, M. H. (1995). *Cities of Light and Heat: Domesticating Gas and Electricity in Urban America*. Penn State University Press. For gas and electricity in Denver and Kansas City.
- Tarr, J. A. and Dupuy, G., eds. (1988). *Technology and the Rise of the Networked City in Europe and America*. Temple University Press.
- Tomory, L. (2012). *Progressive Enlightenment: The Origins of the Gaslight Industry, 1780–1820*. MIT Press. Lower priority, since the 2011 article covers the network years.

Primary sources that would settle the open points

- *Historical Statistics of the United States*, Millennial Edition (Cambridge, subscription): dwellings with electric service by year.
- Gas Light and Coke Company, Minutes of the Court of Directors and of the Committee of Works, London Metropolitan Archives, B/GLCC/1 and B/GLCC/87–88. Tomory's main source.
- Commonwealth Edison Company annual reports, which Platt uses for customer counts.
- Rural Electrification Administration annual reports, 1935 to 1940.
- The 2008 New York City and Verizon cable franchise agreement.
- FCC Form 477 block-level deployment data, which is public and could be mapped directly.
- Smith's Figure 3, a 1903 map of all 22 Chicago stations and substations with street addresses. The address list came through the text scan in garbled form and the map did not. A page image would let the locations be mapped against the street network.